Noise Traders in General Equilibrium

Paro Suh

CU Boulder Economics

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Outline

1. Introduction

2. The Baseline Economy

3. Results

3-1. Numerical Solution

3-2. Comparative Statics

"Noise Traders" in This Paper

Noise Traders are defined as agents with portfolio demand not equal to **FI-RE** demand.

(Full Information - Rational Expectations)

In a market with N risky assets,

- Let $\theta_{i,t} \in \mathbb{R}^N$ denote agent i's (dollar) portfolio demand at time t
- **FI-RE Demand** $\theta_{i,s}^{FI-RE} \in \mathbb{R}^N$ is the portfolio component of the solution to:

$$(C_{i,s}^{FI-RE}, \theta_{i,s}^{FI-RE})_{s \geq t} = \arg \max_{\{C_{i,s}, \theta_{i,s}\}_{s \geq t}} \mathbb{E}_t^{true} \left[U_i \left(\{C_{i,s}\}_{s \geq t} \right) \right],$$

  an EU problem under correct beliefs, standard wealth dynamics and correct return process
- **Noise Traders Demand** deviates from FI-RE demand: $\theta_{i,t} \neq \theta_{i,s}^{FI-RE}$

"Noise Traders" in Literature

- Noise Traders are *broadly* described as agents with trading behavior deviating from "benchmarks"

Category	Benchmark	Deviation Margin	Explanation
Informational	Full Information Rational Expectations (FI-RE)	Information Structure	Agents have distorted expectations about fundamentals.
Constraint-Based	Frictionless Optimization	Feasible Set	Rational agents with complete information, but portfolio demand is distorted by market frictions.
Preference-Based	Standard Expected Utility	Preferences	Agents are fully informed and optimizing, but preferences depart from EU
Reduced-Form Demand Shocks	Demand Fully Derived from Optimization	Derivation of Demand	Asset demand includes an exogenous component not derived from fundamentals nor utility maximization.

references

- In this paper, we focus on **informational distortions** relative to a FI-RE benchmark.

"Noise Traders" in General Equilibrium

Natural question to ask: "*Why* and *how* do agents deviate?"

- A substantial literature is devoted to studying these questions.
- **Not the focus of this paper.**

- Noise traders are *taken as given* in this model with the simplest form of demand distortion
- The focus is, instead, on the **general equilibrium consequences** of their behavior. Specifically:
  1. Deviations from the efficient-market benchmark
     - *How does the equilibrium pricing kernel depart from the FI-RE outcome?*
  2. Wealth distribution dynamics
     - *Do noise traders accumulate or lose wealth in equilibrium?*

Arbitrage Pricing Theory (APT)

- As a reference framework for general equilibrium pricing, we turn to **Arbitrage Pricing Theory:**
  1. Arbitrage-free markets imply the Euler equation $\boxed{\mathbb{E}_t\left[M_{t+dt}R_{k,t+dt}\right] = M_t}$
     (M_t: Stochastic Discount Factor (SDF), $R_{k,t}$: the return diffusion of asset k)
  2. If risk is driven by Brownian shocks $\mathbf{B}_t$, the SDF evolves as $\boxed{\frac{dM_t}{M_t} = -r_t\,dt - \lambda_t^\top d\mathbf{B}_t}$
     (r_t: risk free rate, λ_t : market prices of risk)
  3. Substituting this into the Euler equation yields $\boxed{\mathbb{E}_t\left[R_k - r\right] = \beta_k^\top \lambda}$
     where β_k is the collection of factor loadings on asset k.
- In other words, understanding equilibrium pricing boils down to **understanding the SDF** M_t, whose diffusion coefficients r_t and λ_t are functions of the underlying economic state.
- Relative to FI-RE, noise traders *expand* the state dependence.

Arbitrage Pricing with Noise Traders

- The state vector of a FI-RE economy is $\mathbf{x}_t$ = (fundamentals)
- With noise traders, the equilibrium state vector expands to

$$\mathbf{x}_t = (\text{fundamentals, behavioral states, wealth distribution})$$

  - Behavioral states represent shocks that distort noise traders' asset demands
  - Wealth distribution affects aggregate risk-bearing capacity in equilibrium
- The SDF M_t (due to r_t and λ_t) is a function of the equilibrium state vector

$$M_t = M(\mathbf{x}_t)$$

  - So noise traders do not change the arbitrage structure of the economy; they enlarge the state space that governs the pricing kernel and risk premia

What This Paper Does

- **Model Setup**
  - Continuous-time general equilibrium with perpetual youth
    in simple endowment (Lucas tree) & production (AK) environments
  - Two types of agents (rational & noise), belief distortions captured by a behavioral state
- **Equilibrium Characterization**
  - Derive optimal policies, solve for equilibrium prices and wealth distribution
  - Characterize the equilibrium pricing kernel
- **Quantitative Implementation**
  - Numerical solver for coupled HJB – market clearing system
  - Compute equilibrium objects over the state space, simulate economy

Position in the Literature (1)

- **Behavioral Asset Pricing (Noise Traders)**
  - De Long et al.(1990, 1991), Campbell&Kyle(1993), Barberis&Schleifer(2003)
    - Studies noise trader risk and their effect on prices, typically operating in PE or OLG settings with constant wealth
  - I move to a GE setting, where noise is a state variable affecting the market price of risk
- **Macro-Finance and Wealth Dynamics**
  - Brunnermeir&Sannikov(2014), He&Krishnamurthy(2013)
    - Use financial frictions to show that the wealth distribution is a first-order state variable for the pricing kernel
  - I let (psychological) belief distortions take the place of "financial frictions"

Position in the Literature (2)

1. **Conitnuous-Time GE with Perpetual Youth**
   - Yaari(1965), Blanchard (1985), Dumas(1992)
     - Allows for a stationary wealth distribution, providing a tractable way to model long-run heterogeneity
   - I use this framework to sustain the interaction between the two types of agents.

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Perpetual Youth

- **Demographics**. Continuous time, constant unit mass
  - Poisson death rate λ; Newborn flow λ
- **Annuity Market.**
  - No operative bequests, wealth redistributed to survivors via mortality credits
  - Effective returns $r_t + \lambda$ offsets higher discounting $\rho + \lambda$
- **Government Transfers.** fiscal transfer mechanism preventing permanent zero-wealth
  - $dT_t = \tau Y_t dt \implies W^{birth} = \frac{\tau Y_t}{\lambda}$

Heterogenous Agents

- 2 types of agents: **Noise Traders** $\mathcal{N}$ and **Rational Traders** $\mathcal{R}$:
  - Fixed fraction $\alpha \in [0, 1]$ are noise traders, the remaining $1 - \alpha$ are rational
- $\mathcal{R}$ and $\mathcal{N}$ have different asset demands:
  - Rational demand coincide with FI-RE demand $\theta_{i,t}^{\mathcal{R}} = \theta_t^{FI-RE}$
  - Noise Trader demand deviates from FI-RE demand $\theta_{i,t}^{\mathcal{N}} \neq \theta_t^{FI-RE}$
- Noise Trader demand $\theta_{i,t}^{\mathcal{N}}$ is *distorted* by "Investor Sentiment" (s_t)
  - Sentiment is non-fundamental distortion in subjective beliefs about future asset payoffs

    DeLong et al. (1990, 1991), Barberis, Shleifer & Vishny (1998), Baker & Wugler (2006, 2007), Barberis (2018)
  - In this paper, sentiment enters as a *belief wedge* on return moments, shifting risky demand

Investor Sentiment

- Sentiment s_t evolves according to an Ornstein-Uhlenbeck process:

$$ds_t = \kappa_s(\bar{s} - s_t)dt + \sigma_s dB_t^s$$

  - s_t: aggregate sentiment
  - $\bar{s}$: long-run (average) sentiment
  - $\kappa_s > 0$: speed of mean reversion (persistence of s_t)
  - $\sigma_s > 0$: volatility of sentiment shocks
  - B_t^s : standard Brownian motion (sentiment innovation)

- *Note.* Sentiment Innovation may be correlated with fundamental innovation ($\mathbf{B}_t^F$):

$$dB_t^s dB_t^F = \rho_{Fs} dt, \qquad \rho_{Fs,i} \in (-1, 1)$$

  - In the baseline, we assume orthogonal shocks: $\rho_{Fs} = \mathbf{0}$

Production

- **Endowment Economy (Lucas Tree).** $Y_t = D_t$ with $D_t = \bar{D}\exp(z_t)$
  - Single productive asset in fixed unit supply $\bar{S} = 1$.
- **Fundamental Shocks z_t.**
  1. Level Shocks. (i.i.d. growth) :
  $$dz_t = \mu_z dt + \sigma_z dB_t^z$$
     - Economy is homogeneous of degree one in D_t
  2. Growth Shocks (persistent expected growth):
  $$dz_t = g_t dt + \sigma_{z,0} dB_t^z \qquad dg_t = \kappa_g(\bar{g} - g_t)dt + \sigma_g dB_t^g$$
     - g_t acts as an additional state variable

Financial Assets

1. **Risk-Free Asset** (b_t)

$$\frac{db_t}{b_t} = r_t dt$$

   - Zero-net supply in equilibrium.

2. **Risky Asset.** claims to the Lucas tree

$$\frac{dR_t}{R_t} \equiv \frac{dP_t + D_t dt}{P_t} = \mu_t^R dt + \Sigma_t^R \cdot d\mathbf{B}_t$$

   - Expected return μ_t^R and volatility/loading vector Σ_t^R are determined in equilibrium
   - $d\mathbf{B}_t$ collects the economy's primitive shocks

The Individual's Problem

The expected lifetime utility of agent i, with type $k \in \{\mathcal{N}, \mathcal{R}\}$, born at time t

$$\mathbb{E}_t^k \left[\int_t^\infty e^{-(\rho+\lambda)(s-t)} \frac{(C_i^k)^{1-\gamma_k}}{1-\gamma_k} ds \right]$$

subject to the wealth accumulation equation.

- Agents choose consumption $C_{i,t}^k$ and (dollar) position in risky asset $\theta_{i,t}^k$.
- Parameters. ρ *(impatience)*, λ *(possibility of death)*, γ_k *(type specific risk aversion)*
- $\mathbb{E}_t^k[\cdot]$ represents type specific beliefs
  - Objective beliefs for $\mathcal{R}$, Distorted beliefs for $\mathcal{N}$ (Girsanov Distorted Drift: $\tilde{\mu}_t^{\mathcal{N}} = \mu_t^{\mathcal{R}} + \psi_s s_t$)

Individual Wealth Dynamics

Individual i with type $k \in \{\mathcal{N}, \mathcal{R}\}$ accumulate financial wealth according to:

$$dW_{i,t}^k = \left[(r_t + \lambda)W_{i,t}^k + \theta_{i,t}^k(\mu_t^R - r_t) - C_{i,t}^k\right] dt + \theta_{i,t}^k \Sigma_t^R \cdot d\mathbf{B}_t$$

- Newborn agents start with financial wealth $W_t^{birth} > 0$
- The dollar position $\theta_{i,t}^k$ in the risky asset earns expected return $(\mu_t^r - r_t)$ per dollar invested
- Remaining wealth$(W_{i,t} - \theta_{i,t})$ is invested in the risk-free asset
- Wealth volatility is driven by exposure to aggregate shocks $d\mathbf{B}_t$, with return loading vector Σ_t^R.

The Hamilton-Jacobi-Bellman Equation

The Hamiltonian-Jacobi-Bellman Equation

$$(\rho + \lambda)V^k = \max_{\{C^k, \theta^k\}} \left\{ \frac{(C^k)^{1-\gamma_k}}{1-\gamma_k} + \frac{1}{dt}\mathbb{E}_t^k \left[dV^k \right] \right\}$$

- Homothetic Value Functions: $V^k(W, \mathbf{x}) = \frac{W^{1-\gamma_k}}{1-\gamma_k} v^k(\mathbf{x})$
- Wealth Normalized Controls: Optimal policies scale proportionally with wealth
  $\implies$ the problem reduces to solving $c^k \equiv C^k/W^k$(consumption rate) and $\phi^k \equiv \theta^k/W^k$(wealth share)

First Order Conditions

- The F.O.C. w.r.t. c^k :

$$c^{k,*}(\mathbf{x}) = (v^k(\mathbf{x}))^{-\frac{1}{\gamma_k}}$$

- The F.O.C. w.r.t. ϕ^k :

$$\phi^{k,*}(\mathbf{x}) = \frac{\tilde{\mu}^k(\mathbf{x}) - r(\mathbf{x})}{\gamma_k \left\|\Sigma^R(\mathbf{x})\right\|^2} + \frac{\Sigma^R(\mathbf{x}) \cdot \left(\Sigma^{\mathbf{x}}(\mathbf{x})^\top \nabla \ln v^k(\mathbf{x})\right)}{\gamma_k \left\|\Sigma^R(\mathbf{x})\right\|^2}$$

where $\tilde{\mu}_t^k = \begin{cases} \mu_t^R & \text{(for } k \in \mathcal{R}) \\ \mu_t^R + \psi_s s_t & \text{(for } k \in \mathcal{N}) \end{cases}$

Stochastic Discount Factor and Market Price of Risk

- Agent's marginal utility of wealth: $M_t^k = e^{-(\rho+\lambda)t}(W_t^k)^{-\gamma_k} v^k(\mathbf{x}_t)$

- Applying Itô's Lemma:

$$\frac{dM_t^k}{M_t^k} = -r\,d_t - \Lambda_t^k \cdot d\mathbf{B}_t$$

- **The Market Price of RIsk** Λ_t^k

$$\Lambda_t^k = \gamma_k \phi_t^k (\Sigma_t^R)^\top - (\Sigma^{\mathbf{x}})^\top \nabla \ln v^k(\mathbf{x}_t)$$

Aggregation

- Within each type $k \in \{\mathcal{N}, \mathcal{R}\}$, agents are ex-ante identical and face only aggregate shocks
  - cross-sectional distribution collapses to two representative per-capita wealth levels W_t^k
- It is Convenient to Define
  - Define aggregate wealth for each type: $\mathbf{W}_t^k = \alpha_k W_t^k$, where $\alpha_k = \begin{cases} \alpha & \text{for } k = \mathcal{N} \\ 1-\alpha & \text{for } k = \mathcal{R} \end{cases}$
  - Total Financial Wealth: $\mathbf{W}_t = \mathbf{W}_t^{\mathcal{N}} + \mathbf{W}_t^{\mathcal{R}}$
  - Aggregate wealth share of noise traders ω_t

$$\omega_t \equiv \frac{\mathbf{W}_t^{\mathcal{N}}}{\mathbf{W}_t}$$

Market Clearing Conditions

Due to risk free market clearing, $\mathbf{W}_t = \alpha\theta_t^{\mathcal{N}} + (1-\alpha)\theta_t^{\mathcal{R}}$,

- **Risky Asset MCC**.

$$\omega_t \phi_t^{\mathcal{N}} + (1-\omega_t)\phi_t^{\mathcal{R}} = 1$$

- **Goods MCC**.

$$\omega_t c_t^{\mathcal{N}} + (1-\omega_t)c_t^{\mathcal{R}} = \frac{D_t}{P_t(1-\tau)} = \frac{1}{(1-\tau)q_t}$$

where $q_t \equiv \frac{P_t}{D_t}$

Reduced State Vector

Normalized MCCs imply that equilibrium depend on shares, not levels

$$\mathbf{x}_t = \begin{bmatrix} z_t & s_t & \omega_t \end{bmatrix}^\top$$

- z_t: fundamental productivity (exogenous SDE)
- s_t: belief distortion of noise traders (exogenous SDE)
- ω_t: wealth share of noise traders (endogenous state)

Aggregate Wealth ($\mathbf{W}_t^k$) Dynamics

Let $\mathbf{W}_t^k = \alpha_k W_t^k$, for $k \in \{\mathcal{N}, \mathcal{R}\}$, where $\alpha_{\mathcal{N}} = \alpha$ and $\alpha_{\mathcal{R}} = (1-\alpha)$. Then,

$$d\mathbf{W}_t^k = \mathbf{W}_t^k \mu_t^{W,k} dt - \underbrace{\lambda \mathbf{W}_t^k}_{\text{exit term}} dt + \underbrace{\alpha_k \left(\tau D_t \bar{S}\right)}_{\text{inflow term}} dt + (\text{diffusion term})$$

where the drift of financial wealth:

$$\mu_t^{W,k} = r_t + \phi_t^k (\mu_t^{risky} - r_t) - c_t^k$$

Wealth Share (ω_t) Dynamics

Applying Itô's Lemma to $\omega_t \equiv \mathbf{W}_t^{\mathcal{N}}/\mathbf{W}_t$ yields $d\omega_t = \mu_t^{\omega} dt + \Sigma_t^{\omega} \cdot d\mathbf{B}_t$ with

$$\mu_t^{\omega} = \omega_t(1-\omega_t)\left(\mu_t^{W,\mathcal{N}} - \mu_t^{W,\mathcal{R}} - \Gamma_t\right) + \underbrace{\frac{\tau}{1-\tau} q_t(\alpha - \omega_t)}_{\text{Stabilization}}$$

- Γ is the Itô correction term, and

$$\mu_t^{W,\mathcal{N}} - \mu_t^{W,\mathcal{R}} = (\phi_t^{\mathcal{N}} - \phi_t^{\mathcal{R}})(\mu_t^{risky} - r_t) - (c_t^{\mathcal{N}} - c_t^{\mathcal{R}})$$

Normalized Hamilton-Jacobi-Bellman

The normalized HJB

$$(\rho+\lambda)v^k(\mathbf{x}) = \max_{c^k,\phi^k}\left\{(c^k)^{1-\gamma} + (1-\gamma)v^k\mu_W^k - \frac{\gamma(1-\gamma)}{2}v^k\left\|\phi^k\Sigma_R\right\|^2 \right.$$
$$\left. +\mathcal{L}_{\mathbf{x}}v^k + (1-\gamma)\phi^k\Sigma_R\cdot\left(\Sigma_{\mathbf{x}}^\top\nabla v^k\right)\right\}$$

- $\mathcal{L}_{\mathbf{x}}$ is the generator for the state process $\mathbf{x}$ alone

$$\mathcal{L}_{\mathbf{x}}v^k = \nabla v^k\cdot\mu_{\mathbf{x}} + \frac{1}{2}\mathrm{tr}\left(\Sigma_{\mathbf{x}}\Sigma_{\mathbf{x}}^\top\nabla^2 v^k\right)$$

First Order Conditions

1. Optimal consumption-wealth ratio

$$c^k(\mathbf{x}) = \left(v^k(\mathbf{x})\right)^{-\frac{1}{\gamma}}, \qquad C^k(\mathbf{x}) = v^k(\mathbf{x})^{-\frac{1}{\gamma}} W$$

- $v(x)^{-\frac{1}{\gamma}}$ is the *propensity to consume (MPC) out of Wealth*

2. Optimal Portfolio Share

$$\phi^k(\mathbf{x}) = .\frac{\tilde{\mu}^k(\mathbf{x}) - r(\mathbf{x})}{\gamma \left\|\Sigma_R(\mathbf{x})\right\|^2} + \frac{\Sigma_R(\mathbf{x}) \cdot \left(\Sigma_{\mathbf{x}}(\mathbf{x})^\top \nabla \ln v^k(\mathbf{x})\right)}{\gamma \left\|\Sigma_R(\mathbf{x})\right\|^2}$$

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Numerical Implementation

On grid for $\mathbf{x}_t = \begin{bmatrix} z_t & s_t & \omega_t \end{bmatrix}^\top$, we solve for $\{q(\mathbf{x}), r(\mathbf{x}), v^{\mathcal{N}}(\mathbf{x}), v^{\mathcal{R}}(\mathbf{x})\}$

- **Step 1.** GE Mapping iterations (coarse fixed point)
  - **Step 1-1.** update prices and generator $(\mu_\omega, \Sigma_\omega)$ given policies
  - **Step 1-2.** coarsly solve for HJB (small iterations) and derive policies
  - repeat 1-1, 1-2 untill $\|q_{new} - q_{old}\|, \|r_{new} - r_{old}\|$ are stable
- **Step 2.** Fix q^*, r^* derived from **Step 1** and solve HJBs to convergence (tol= 10^{-6})
  - damped pseudo-Newton / policy-evaluation with backtracking line search
- **Step 3.** Use HJB derived from **Step 2** to update prices q^{**}, r^{**}
  - Verify $q^{**} \approx q^*$ and $r^{**} \approx r^*$

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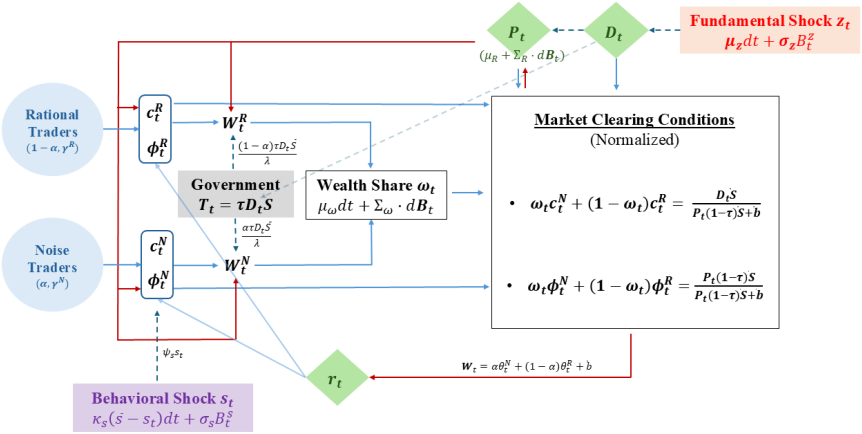
Model Depiction



Dividend Growth Process

Fundamental state z_t follows

$$dz_t = g_z dt + \sigma_z dB_t^z, \quad |z_t| \leq z_{max}$$

- why? long-run dividend data (Shiller; S&P) show high persistence, mean reversion is weak
- $g_z = 0.016$ (average real dividend growth) and $\sigma_z = 0.05$ (growth volatility)
- z_{max} covers three standard deviations of long-run cumulative growth

$$z_{max} \leftarrow \frac{3\sigma_z}{g_z} \approx \frac{3 \times 0.05}{0.016} \approx 0.75$$

Noise Trader References

- **Informational Deviations.** Belief Distortions, Non-Bayesian Updating, Incomplete Information

  DeLong et al. (1990), Bordalo, Gennaioli & Shleifer (2018), Barberis et al. (2015)

- **Constraint Based.** Short Sale, Capital Requirements

  Brunnermeier&Pedersen (2009), He&Krishnamurthy (20130)

- **Preference Based.** Prospect Theory, Habit Formation, Ambiguity Aversion

  Benartzi&Tahler(1995), Campbell&Cochrane(1999), Barberis, Huang & Santos (2001)

- **Reduced Form Demand Shocks.**

  Black (1986), Gabaix&Koijen(2021),

return